

Test 20: TKNN Chern number C?=1 (gapped anchors)

Verdict: PASS | Generated: 2026-09-02 15:30:18 | Suite: AB-Cloud v19 (Julia 1.12.0)

What this test verifies

EXACT: FHS C?=+1 on the MAGNETIC BZ (q $\times$ Harper; anchors ?=1/4,1/3,1/5); ?=1/2 = Dirac touching, per-band C undefined ? touching-ladder rungs need L ? 0 (mod 4) so (?/2,

METHOD: EXACT: FHS C?=+1 on the MAGNETIC BZ (q $\times$ Harper; anchors ?=1/4,1/3,1/5); ?=1/2 = Dirac touching, per-band C undefined ? touching-ladder rungs need L ? 0 (mod

Test 20 HARDCORE: anchors 1/4,1/3,1/5 ? +1 ?; ladder 12..72 ? 1,1,1,1,1,1; ?=1/2 touching ?L, mass branch 0 ?.

Result [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

- 'plots/plot_01' (.png 600dpi / .svg / .pdf)

Plots

- 'plots/plot_02' (.png 600dpi / .svg / .pdf)

- 'plots/plot_03' (.png 600dpi / .svg / .pdf)

- 'plots/plot_04' (.png 600dpi / .svg / .pdf)

- 'plots/plot_05' (.png 600dpi / .svg / .pdf)

- 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: julia ab_cloud_v19.jl --test 20 --no-two-pass

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

Independent verification guide

(no log_comp entries)

FULL COMPUTATION LOG

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TEST 20 (pass 2, HARDCORE): TKNN Chern number ? deep audit

CONSOLE CAPTURE

Anchors {1/4, 1/3, 1/5} + size ladder 12?72 + ?=1/2 touching audit

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?=1/4: C? = +1 (12-pt grid) ? +1 (24-pt grid) ?, min gap 1.646

?=1/3: C? = +1 (12-pt grid) ? +1 (24-pt grid) ?, min gap 1.268

?=1/5: C? = +1 (12-pt grid) ? +1 (24-pt grid) ?, min gap 1.554

?=1/3 ladder L=12 (48 k-points): C? = +1, min gap 1.268 ?

?=1/3 ladder L=18 (108 k-points): C? = +1, min gap 1.268 ?

?=1/3 ladder L=24 (192 k-points): C? = +1, min gap 1.268 ?

?=1/3 ladder L=36 (432 k-points): C? = +1, min gap 1.268 ?

?=1/3 ladder L=48 (768 k-points): C? = +1, min gap 1.268 ?

?=1/3 ladder L=72 (1728 k-points): C? = +1, min gap 1.268 ?

?=1/2 L=12: bare min gap = 5.48e-16 (touching), singular links 0

?=1/2 L=16: bare min gap = 5.48e-16 (touching), singular links 0

?=1/2 L=24: bare min gap = 5.48e-16 (touching), singular links 0

?=1/2 L=48: bare min gap = 5.48e-16 (touching), singular links 0

?=1/2 L=72: bare min gap = 5.48e-16 (touching), singular links 0

?=1/2 mass branch (24x12 grid): C(m=+0.3) = +0 (gap 0.435), C(m=?0.3) = +0 (gap 0.435) ?

??

anchors C?=+1 {1/4,1/3,1/5} PASS ? TKNN Diophantine t?=+1, grid-converged 12?24 points

ladder C?=+1, L=12..72 PASS ? magnetic-BZ FHS on L = 12/18/24/36/48/72 (?=1/3)

?=1/2 Dirac touching + branch PASS ? bare gap ? 1e-16 ?L (rungs L ? 0 mod 4 ? Dirac points on-grid); mass-regularized C? = 0 (chirality-balanced)

Test 20b [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS